NOTICE

THIRD YEAR B.TECH. COMPUTER SCIENCE AND ENGINEERING

MACHINE LEARNING - INDIVIDUAL PROJECT

Academic Year: 2026 27

All TY CSE students are hereby informed that an **Individual Machine Learning Project** has been assigned to assess their practical understanding and implementation skills developed through the **Machine Learning subject**.

The project has been assigned to each student **on the basis of their Roll Number**.

Students must complete the project individually and demonstrate the complete Machine Learning pipeline from **problem definition to deployment.**

Project Requirements

Each student must complete the following stages:

1. Problem Definition

- Clearly define the selected problem.
- Explain the real-world requirement.
- Identify the input and expected output.
- Identify whether the problem is classification, regression, clustering, recommendation, or another suitable ML task.

2. Aim and Objectives

Clearly define:

- Aim of the project.
- Specific objectives to be achieved.

3. Dataset Collection

Students must:

- Find a relevant dataset from a reliable online source, **or**
- Create their own dataset where appropriate.
- Provide the dataset source/location.
- Explain the features and target variable.
- Analyze the size, structure and characteristics of the dataset.

4. Data Preprocessing

Students must identify and apply appropriate data preprocessing techniques, wherever required, such as:

- Handling missing values
- Removing duplicate records
- Handling outliers
- Encoding categorical data

- Feature scaling/normalization
- Feature selection
- Other appropriate preprocessing techniques

Students must **justify why each preprocessing technique was required and how it affected the data**.

5. Exploratory Data Analysis and Visualization

Students must analyze the dataset using suitable statistical methods and visualizations.

Examples include:

- Distribution plots
- Scatter plots
- Bar charts
- Box plots
- Correlation heatmap
- Confusion matrix
- Decision boundary
- Actual vs predicted plot
- Other relevant visualizations

Students must explain the observations obtained from each important visualization.

6. Feature and Target Selection

Students must clearly identify:

- Input features (X)
- Target/output variable (y)

The selection of important features must be explained and justified.

7. Train-Test Split

Students must divide the dataset into appropriate training and testing data and explain:

- Why the dataset was divided
- Training data
- Testing data
- Selected train-test ratio
- Role of random_state, wherever applicable

8. Model Selection and Algorithm Comparison

Students must implement **at least two suitable Machine Learning algorithms** for their selected problem.

The performance of the different algorithms must be compared using appropriate evaluation metrics.

Students must prepare a comparison such as:

Algorithm	**Evaluation Metric(s)**	**Result**
Algorithm 1	_____	_____
Algorithm 2	_____	_____
Algorithm 3, if applicable	_____	_____

Students must **justify the selection of the final algorithm based on the experimental results and characteristics of the problem**.

9. Model Training

Students must train the selected Machine Learning model using the training dataset.

They must understand and explain:

- How the model is trained
- What the model learns from the training data
- Important parameters used during training

10. Model Evaluation

Students must evaluate the trained model using appropriate evaluation metrics.

Examples:

Classification:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC, wherever applicable

Regression:

- MAE
- MSE
- RMSE
- R^2

Students must interpret the obtained results rather than simply reporting the numerical values.

11. Model Improvement

Students must perform suitable experiments to improve or analyze the model.

Examples:

- Changing algorithm parameters
- Feature selection
- Changing train-test ratio
- Scaling comparison
- Hyperparameter tuning
- Comparing different algorithms
- Handling overfitting/underfitting

The effect of each important modification must be recorded and explained.

12. Final Model Selection

Students must select the final model based on experimental results and provide a proper justification for the selection.

The selection should be supported by **results, evaluation metrics and problem requirements**, not merely by stating that one algorithm is "better."

13. Front-End Development

Students must develop a simple user interface for their ML application.

The front end should allow the user to:

Enter/Input Data → Submit Data → Generate Prediction → Display Output

For example:

House Price Prediction

Enter Area: 1500 sq.ft.

Enter Bedrooms: 3

Enter Bathrooms: 2

Enter Parking: 1

[Predict Price] Predicted House Price: ₹68,50,000

Students may use a suitable technology such as **Streamlit** or another appropriate framework.

14. Model Deployment

The complete project must be properly organized and uploaded to the student's **GitHub profile/repository** along with **project report :**

The repository should contain, wherever applicable:

Project/

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├── Dataset/

├── Notebook/

├── Source Code/

├── Trained Model/

├── Front End/

├── Requirements.txt

├── README.md

└── Project Documentation

The README.md file should contain:

- Project title
- Aim
- Objectives
- Problem statement
- Dataset information
- ML pipeline
- Algorithms used
- Experimental results
- Final model
- Front-end details
- Screenshots
- Installation/execution instructions
- GitHub repository link

Expected Viva and Demonstration

Each student must demonstrate the **complete working project individually**.

During the viva, students will be required to justify and explain **each stage of the Machine Learning pipeline**, including:

Problem Definition → Dataset → Data Understanding → Preprocessing → EDA → Feature Selection → Train-Test Split → Algorithm Selection → Model Training → Prediction → Evaluation → Visualization → Model Improvement → Final Model → Front End → Deployment

At the time of viva examiner may also modify parameters, remove a feature, change an algorithm, or ask the student to execute an additional experiment.

Students must be able to explain the expected effect and demonstrate their understanding.

Students should therefore **NOT JUST COPY CODE FROM ONLINE SOURCES OR GENERATE A PROJECT WITHOUT UNDERSTANDING ITS IMPLEMENTATION**.

Every important line, technique, algorithm, result and design decision should be understood by the student.

EXPECTED LEARNING OUTCOME

The purpose of this project is to ensure that every student develops the ability to independently:

Understand a real-world problem → Prepare data → Analyze data → Select an appropriate ML algorithm → Train a model → Evaluate and improve it → Build an application → Deploy it → Explain and defend the complete solution.

Successful completion of this project will help students develop a **complete practical understanding of the Machine Learning pipeline** and apply the concepts learned through all practicals to a real-world problem.

Project Assignment: Based on Roll Number

Project Type: Individual

Submission: GitHub Repository + Project Demonstration + Viva

Evaluation: Implementation, Results, Understanding, Justification, Viva and Deployment